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Book Review

Mixed-Effects Models in S and S-Plus

J.C. Pinheiro and D.M. Bates. New York: Springer Verlag, 2000. ISBN 0-387-98957-9. x+528 pp.

Well-designed computing tools have played a fundamental role in the creation and sustenance of widespread interest in the application of statistical models involving fixed and random effects. This book (referred to as MEMSS here and at the authors' web site) and the family of S-PLUS/R procedures at its core (`lme` for linear mixed effects (LME) models, and `nlme` for nonlinear mixed effects (NLME) models) contribute strongly to the growth and refinement of effective understanding and use of mixed effects models by providing

- a graduated, graphics-rich introduction with detailed transcripts of program behavior on a wide variety of datasets provided with the software,
- an intuitive and flexible language for describing mixed-effects models and serial dependence structures for use in exploratory plotting, model formulation, fitting, and diagnosis,
- self-contained expositions of theoretical background and computational details of both linear and nonlinear mixed effects models,
- tools to simplify simulation and initialization of complex models,
- supplementary documents describing correspondences between nlme/lme and SAS(tm) PROC MIXED, and
- a web site (`http://nlme.stat.wisc.edu`) including scripts of all command sequences used in MEMSS, along with a mailing list archive to help users track software evolution and user-community/developer dialogues.

The computing and documentation resources made available *gratis* by Pinheiro and Bates are substantial and highly self-sufficient contributions to the toolkit of every S-Plus- or R-compatible working statistician. Nevertheless, MEMSS should be welcomed by all users of nlme and S-Plus/R as a panoramic and highly interactive guide to a large and growing family of statistical methods and algorithms of great practical interest.

Students of mixed effects models have enjoyed access to a number of useful monographs on the subject, and the present volume defers (pp. 57, 305) to Searle, Casella and McCulloch (1992) and Vonesh and Chinchilli (1997) for comprehensive treatments of the theory of LME models, and to Davidian and Giltinan (1995) and Vonesh/Chinchilli for the theory of the NLME model. However, the user of MEMSS will encounter numerous opportunities for theoretical enlightenment. About 17% of the text (excluding appendices) is explicitly devoted by chapter heading to formal and theoretical considerations, and additional material of a theoretical nature crops up throughout the remainder of the book, e.g., in a section about extending the standard LME model to accommodate serial or spatial correlation.

The style of the book is highly conversational and "interactive", in the sense that the text is frequently punctuated with S-Plus/R dialog excerpts that a user can execute (with improvised variations) while reading. Thus the book begins with analysis of a one-way layout, emphasizing well-designed graphics for the data and model residuals as the analysis passes from a simple linear to a random intercept model. The first chapter continues with examples of randomized blocks, unbalanced designs, random interactions, growth curves, and a pair of models illustrating nested random effects. The introductory treatments, occupying about 50 pages, are brief and successfully motivational, with minimal theory. In contrast, the volume of Vonesh and Chinchilli (1997) begins with 35 pages of preliminaries on matrix algebra and general statistical theory; Davidian and Giltinan (1995) are focused on NLME models and begin with narratives and data graphics on examples in pharmacokinetics, and move very rapidly into formalisms to expose the wide variety of NLME models that have been proposed. These different approaches to the subject at hand complement one another. In different situations, the working statistician will sometimes find one volume more immediately helpful than the others.

After the introduction to the basics of using `lme`, the underlying theory is developed with an emphasis on computational details, often with links back to the introductory examples. Maximum likelihood (ML) estimation, profiling, restricted ML (REML) and likelihood ratio tests are introduced using the introductory examples. The QR decomposition and re-expression of the random effects covariance in terms of relative precision factors (themselves further re-expressed using matrix logarithm transformations) are worked through in some detail as they are central to evaluation and unconstrained optimization of the (restricted) likelihood. Considerable attention is given to the problem of likelihood-ratio testing for the existence of random effects, with special simulation software introduced to evaluate adjusted critical values proposed by Stram and Lee (1994). This is a good illustration of the utility of well-designed, interoperable software components for probing the interface between theory and application.

The theoretical chapter on LME is followed by "Describing the Structure of Grouped Data". At the outset a syntax for expressions of clustered structures among (scalar) responses is reviewed, using the formula schema `response ~ primary | grouping`. Grouping expressions of the form `g/h/i` denote that `i` is nested within `h`, which is nested within `g`. Thus in a study of repeatedly observed left and right lymph nodes of dogs, the formula `Pixels ~ day | Dog/Side` expresses the nesting of sides within each dog, and identifies `Dog` as the index for independent contributions on `Pixels`. The authors suggest that this class of formulas be regarded as "display formulas", because generic plot functions applied to such formulas produce highly intuitive trellis graphics on the data. Many examples of effective trellis displays are provided throughout the book. One idiosyncrasy of the trellis approach impedes full acceptance of the default approach: the tendency to use colored backgrounds in "labeling strips" often severely obscures labeling text. See Figure 4.30 for an example of the problem.

The interaction between grouping formulae, trellis graphics, and high-level modeling provides a nice example of the effectiveness of object-oriented software environments, and exploits the important capacity for "computing on the language" that is so central to S/S-Plus. It is mildly regrettable that none of the details of this fundamental aspect of the architecture and success of the NLME library are presented in MEMSS. While it is clear that information on these details is peripheral to the *use* of NLME, effective *extension* of NLME by the user community will likely require enrichment of the grouping and modeling sublanguages implied by the software. The statistical computing community leadership needs to expose larger audiences of students and investigators to the details of "computing on the language" in order to foster the growth of methods like these.

The first part of the book, devoted to LME, concludes with two chapters covering fitting and diagnosis of LME models, and extensions to LME for handling heteroscedasticity or embedded serial or spatial correlations. Chapter 5, in which LME models are generalized to accommodate heterogeneous variances, is likely to be of considerable interest to practitioners. The `varPower` and `varConstPower` functions allow the response variance to be a power function of covariates (the default is a power function of the marginal mean). This is followed by further generalization to models with correlations between observations. The special correlation structures include well known patterned structures and spatial correlation structures (similar to those in SAS PROC MIXED). The chapter concludes with discussion of the `gls` function for generalized least-squares estimation. This is an alternative way of specifying the multivariate model for the response through direct modeling of the marginal variance-covariance structure. This may be desirable when the interest is focused not on subject-specific (random) components but rather on population effects. All inference and model selection procedures are likelihood-based, and while there is considerable attention devoted to model diagnostics, some discussion of robustification would be in order.

The second part of the book is devoted to NLME, beginning with an overview of NLME models, continuing with a thirty page excursus of theory and computing for NLME, and concluding with a finale on NLME usage including a brief treatment of the important topic of “self-starting” nonlinear regression modeling tools.

A key issue in NLME model estimation is how the likelihood is approximated. The `nlme` function uses the method of Lindstrom & Bates (1990) which alternates between the `nls` for nonlinear regression and `lme` for random effects modeling. In chapter 7, three different approximation methods to the likelihood function are introduced. Plots and updates of models are illustrated with real examples. An extended nonlinear regression model is also introduced to allow a general covariance structure between observations (within each group level) including heterogeneous variances. This generalization is in line with the generalization in linear models. This generalized least-squares method, despite its reliance on optimization of the Gaussian likelihood, provides consistent estimates as long as the mean function f is correctly specified. It thus has a close connection with the GEE method.

In pharmacokinetical research, NONMEM is a FORTRAN 77 program designed to fit mixed-effects regression models. The approximation is based on the first-order Taylor series expansion at 0 (for the random effects). This option is also available in recently developed SAS procedure NLINMIX. This SAS procedure also allows the method implemented in nlme (second-order approximation) and the facilities in gls. The procedure MIXED in SAS is similar to lme and NLINMIX is similar to `nlme`. The variety of covariance structures implemented in MIXED is passed to NLINMIX.

A few minor comments on some nonlinear regression models may be worth noting here. The “asymptotic regression model” (e.g., p. 511) is derived under the assumption that the growth rate is proportional to the size remaining. This or a cubic transformed version is known to as the von Bertalanffy curve (von Bertalanffy, 1938) which is ubiquitous in fisheries growth literature and other biological sciences. A general power or a logarithmic transformation of this model leads to the Richards curve and or Gompertz curve respectively (Seber & Wild, 1989).

This book will be useful for self-guided research of for an applied course on random effects modeling in statistics or biostatistics. Useful appendices provide references and background information on 31 example datasets, 58 pages of S function documentation, and a graphics-rich discussion of self-starting nonlinear regression methods. Most of the chapters conclude with problem sets, many of which take the form of supervised elaborations of modeling exercises using example data supplied with the software.

Overall, MEMSS sets an extremely high standard for texts on statistical computing subenvironments. The narrative is generally very clear and focused. There is even a sense of tension and adventure in connection with the analysis of manufacturing data on analog MOS circuits, proceeding from the modest plots of Figures 3.11-3.12, through smoother-based criticism of the associated LME in section 4.3, to the satisfying trigonometric NLME denouement at the end of section 8.2. Occasional awkwardnesses (e.g., the vague advice on p. 27 concerning the use of wide confidence intervals on variance components to identify model misspecification, the underqualified remark on p. 273 that nonlinear regression models possess "validity beyond the observed range of the data") are heavily outweighed by the excellent design and thorough treatment of the methods and software. One major methodological concern that deserves an airing but does not seem to get any is the problem of overfitting, or, from the other direction, the problem of properly calibrating inferences subsequent to data-dependent model construction. The authors take us from the exploration of clustering using lists of cluster-specific models through the tailoring of variance components models using various appraisals of model adequacy, but do not consider how this process may affect the validity of ultimate statements of "statistical significance". Such statements are not a focus of any of the analyses presented in the book, but they surely will be for some of the readers, and some acknowledgment of this difficulty is in order, even in a text emphasizing computation. A less pressing but somewhat interesting limitation is the implicit requirement imposed by all the methods presented in the book that responses and covariates be measured at the same times. See `http://alpha.luc.ac.be/~ jlindsey/ms/model.ps` for critical consideration of this issue.

Perhaps the only major caveat one is faced with in contemplating the purchase of this book is its expected longevity in the face of continual technological upheaval. It is clear that the software has already grown beyond the functionality described in MEMSS, with the announcement of version 3.3 and its enhancements of matrix classes for handling models with crossed random effects or heterogeneous spatial clustering. MEM methodology is also evolving, e.g., towards relaxation of the assumption that random effects are Gaussian (see `http://www.bio.ri.ccf.org/Resume/Pages/Ishwaran/glmm.ps` for up-to-date references and a novel approach.) Our opinion is that concerns about possible obsolescence of the text are misplaced. As long as the user keeps up with details of evolution of methods and code by monitoring the web site, it seems to me that the book's usefulness will endure, both as a starting point and as a guide to the active researcher.

In conclusion, we highly recommend this text to statisticians who are interested in using state of the art computing environments and algorithms to analyze grouped and clustered data. We should be particularly grateful to the authors not only for their contributions to the statistical methods, but also for the example they set on how implementations of complex algorithms can be designed, documented and disseminated for the use of a very broad spectrum of investigators.

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